Learning Objectives

- To learn about the engine emissions along with its impact on humans and environment.
- To learn about the engine firing procedure.

6.0 INTRODUCTION

In the earth, all the living organisms, animals and human beings are living with the help of oxygen present in nature (or) atmosphere. For producing power and food they use energy obtained from burning the fuel. Human beings use different fuels to get their heat energy according to their needs. In the same way they use heat energy obtained from burning the fuel for operating their automobiles. Human beings use different approaches for burning the fuels to produce energy or power. For attaining such strategies, they using separate systems for transferring fuel from the fuel tank in to combustion chamber. The system used to perform this function is called as the intake system. The exhaust gases produced from the combustion of fuel inside the combustion chamber are sent out of the engine by another system called as the exhaust system.

6.1 EFFECTS OF POLLUTANTS

The exhaust emissions released from the engine combustion chamber are toxic to the human beings and highly pollute the environment. Hence it is advisable to produce less emission from the engines. It can be achieved by burning the fuels completely in the engine combustion chamber. By controlling the pollutants from the automotive engines, the environment can be maintained as clean and all the human beings, animals and all the plants can live peacefully without any diseases. The following are the important pollutant emissions coming out from the internal combustion engines.

1. Carbon Monoxide
2. Nitric Oxide
3. Hydro Carbon
4. Smoke
5. Particulate (solid, liquid pollution)
6. Sulfur Dioxide and etc.,

The following picture and table present the effects of the above pollutants on human beings

In order to reduce the formation of the above hazardous pollutants the fuel must be burnt completely inside the

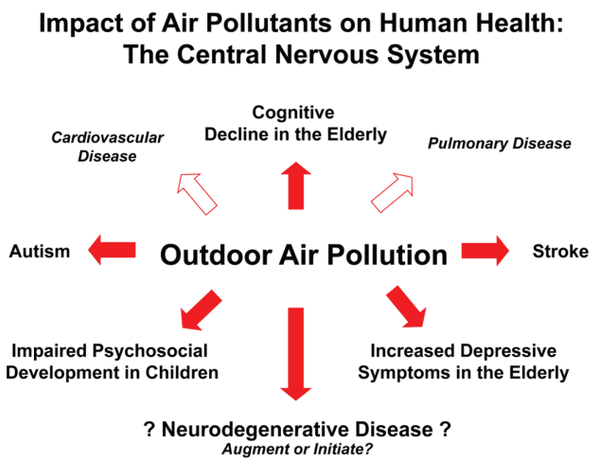


Figure 6.1

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